C9.3 Role of the interconnect

The interconnect can pass or fail an Exclusive Store transaction. A pass indicates that the transaction has been propagated to other cacheable masters. A fail indicates that the transaction has not been propagated to other masters and therefore the Exclusive Store cannot pass.

The interconnect is required to have a monitor, that is used to ensure that an Exclusive Store transaction from a master is only successful if that master could not have received a snoop transaction relating to an Exclusive Store to the same address from another master after it issued its own Exclusive Store transaction. This monitor is referred to as the *PoS exclusive monitor* and it exists within the interconnect at the point of serialization.

C9.3.1 Minimum PoS Exclusive Monitor

The minimum requirement of PoS exclusive monitor is to record when any master performs a shareable transaction related to an Exclusive sequence. The shareable transactions related to an Exclusive sequence are:

- ReadClean with **ARLOCK** asserted
- ReadShared with **ARLOCK** asserted
- CleanUnqiue with **ARLOCK** asserted.

If a master has performed a transaction related to an Exclusive sequence and it then performs an Exclusive Store transaction before a successful Exclusive Store transaction from another master is scheduled, then the Exclusive Store transaction must be successful.

The monitor must support the parallel monitoring of all Exclusive-capable masters in the system.

When the interconnect receives a transaction associated with an Exclusive Load or an Exclusive Store, the monitor registers that the master is attempting an Exclusive sequence. For an Exclusive Store transaction, this must be recorded if the Exclusive Store is failed, i.e. it receives an OKAY response. If the Exclusive Store transaction is successful it is recommended, but not required, that the monitor registers that the master is attempting an Exclusive sequence.

If an Exclusive Store transaction from one master passes, the registered attempts of all other masters is reset. The other masters can only register a new Exclusive sequence after the **RACK** is observed for the Exclusive Store transaction that passed.

When the interconnect receives an Exclusive Store transaction:

- If the PoS exclusive monitor has registered that the master is performing an Exclusive sequence, that is, it has not been reset by another master's Exclusive Store transaction, then the Exclusive Store transaction is successful and is allowed to proceed. An EXOKAY response is returned to the issuing master.
- If the PoS exclusive monitor has not registered that the master is performing an Exclusive sequence, that is, it has been reset by another master's Exclusive Store transaction, then the Exclusive Store transaction is failed and is not allowed to proceed. An OKAY response is returned to the issuing master.

Note

A successful Exclusive Store transaction from a master does not have to reset that the master is performing an Exclusive sequence. The master can continue to perform a sequence of Exclusive Store transactions that will all be successful, until another master performs a successful Exclusive Store transaction.

From reset, the first master to perform an Exclusive Store transaction can be successful, but is not required to be. At that point, all other masters must then register the start of their Exclusive sequence for their Exclusive Store transaction to be successful.

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C9.3.2 Additional address comparison

The interconnect monitor can be enhanced to include some address comparison. A full address comparison is not required and it is permitted to only record a subset of address bits. This approach reduces the chances of an Exclusive Store transaction failing because of another master's Exclusive Store transaction to a different address location. The number of bits of address comparison used is an implementation choice.

Where additional address comparison monitor is used, the monitored address bits are recorded at the start of an Exclusive sequence on either a Load Exclusive or Store Exclusive transaction. It is reset by a successful Store Exclusive transaction from another master to a matching address.

A monitor with additional address comparison must include a minimum monitor of a single bit for every Exclusive-capable master to ensure forward progress.

An Exclusive Store transaction is allowed to progress if one of the following occurs:

- the address monitor has registered an Exclusive sequence for a matching address from the same master and has not been reset by an Exclusive Store transaction from a different master with a matching address
- the minimum single bit monitor has registered an Exclusive sequence from the same master, and it has not been reset by an Exclusive Store transaction from a different master to any address.

Note

In the above description, the term *matching address* is used to describe where a monitor only records a subset of address bits. A matching address is where the address bits that are recorded are identical, but the address bits that are not recorded can be different.

An implementation does not require address monitor for each Exclusive-capable master. Because the address monitor provides a performance enhancement it is acceptable to have fewer address monitors and for the use of these to be IMPLEMENTATION DEFINED. Examples of how the additional address monitors can be used include use on a first-come first-served basis, or allocation to particular masters, or a more complex algorithm can be used.

Additional PoS Exclusive Monitor functionality can be provided to prevent interference, or denial of service, caused by one agent in the system issuing a large number of Exclusive access transactions. This specification recommends that secure exclusive accesses are permitted to make forward progress independently of the progress of non-secure exclusive accesses.

C9.3.3 Multiple interconnect PoS monitors

When the interconnect contains multiple points of serialization, as the serialization for different address ranges is done at different points within the interconnect, then a PoS exclusive monitor can be located at each point of serialization.

C9.3.4 PoS Exclusive Monitor behavior

The following terms are used to describe the behavior of a PoS exclusive monitor:

True pass Describes the case where an Exclusive Store transaction is permitted to progress and when the transaction completes the Exclusive Store will also pass, permitting the master to make forward progress.

True fail Describes the case where an Exclusive Store transaction is failed because another master has already performed a successful Exclusive Store transaction to the same address location, so at least one agent has made forward progress.

False pass Can occur for an Exclusive Store transaction, for which the Exclusive Store will eventually fail. This can only occur when a non-exclusive store transaction from another agent to the same location has occurred. This non-exclusive store transaction from another agent is considered as forward progress for that other agent.

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False fail Can occur for a Exclusive Store transaction in the following circumstances:

- No transaction was issued for the Load Exclusive. This is resolved by re-issuing the Exclusive Store.
- An Exclusive Store transaction from another agent to a different location has been successful between the point that the Exclusive sequence is first registered and the point that the Exclusive Store transaction is scheduled. This is resolved by re-issuing the Exclusive Store.
- An Exclusive Store transaction from another agent to the same location has been successful, but that other agent is destined to eventually fail because a third party has performed a non-exclusive store transaction. This non-exclusive store transaction from the third party is considered as forward progress for that third party.

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